

Powering Progress: How Economic Development Shapes Electricity Outcomes Across Africa

Eqraa Khan, Shutong Zhao, Yining Weng, Yiren Ding | Group 0106-E

2026-03-26

Introduction

Background: Clean Air Task Force (CATF) is a nonprofit organization dedicated to advancing clean energy and climate solutions. It is supporting the development of an Electricity Sector Performance Scorecard for African countries.

Overarching Theme: How does economic development shape electricity access, reliability, and sustainability across African countries?

Research Goal: Identify patterns to inform which indicators CATF should prioritize in the Scorecard.

Target population: African countries with available utility and national electricity data.

Audience: CATF Africa staff and policymakers, identify which country groups underperform and where Scorecard indicators should focus.

Key Variables:

- **GDP** : GDP per capita (USD), measure of economic development
- **Electrification Rate** : % of population with electricity access
- **Urban Access Electricity Rate** : % of urban population with electricity
- **Rural Access Electricity Rate** : % of rural population with electricity
- **Poverty Rate** : % of population below the poverty line
- **SAIDI** : average outage hours per customer per year
- **Urbanization Rate** : % of population living in urban areas
- **Electric Power Consumption** : average kWh consumed per capita

Data Wrangling

- Collapsed multiple rows per country using median (numeric) and mode (categorical)
- Removed implausible values (electrification rate more than 100%, GDP more than \$50,000 , population less than 500,000)
- **GDP group:** countries were grouped into **Low, Middle, and High GDP** categories, with about one-third of countries in each group.
- **Income Group:** Low, Lower-Middle, Upper-Middle, and High using GDP per capita **thresholds**. Low (less than \$1,000) / Lower-Middle (\$1,000 -\$4,000) / Upper-Middle (\$4,000 - \$12,500) / High (greater than \$12,500)
- **Urban Rural Gap:** urban minus rural access rate (percentage points)
- **Poverty Group:** Low , Middle , High groups, with about one-third of countries in each group, with about one-third of countries in each group.

Question #1: (Yining) - How is GDP per capita associated with electrification rate across African countries?

How is GDP per capita associated with electrification rate across African countries?

Relevance:

Electrification rate is one of the most fundamental measures of energy access. Examining it against GDP helps identify whether economic development alone drives access, and which countries fall below expectations for their income level – the clearest targets for CATF intervention.

Q1: Variables & Approach

Variables used:

- **GDP** : GDP per capita (USD)
- **Electrification Rate** : % of population with electricity access

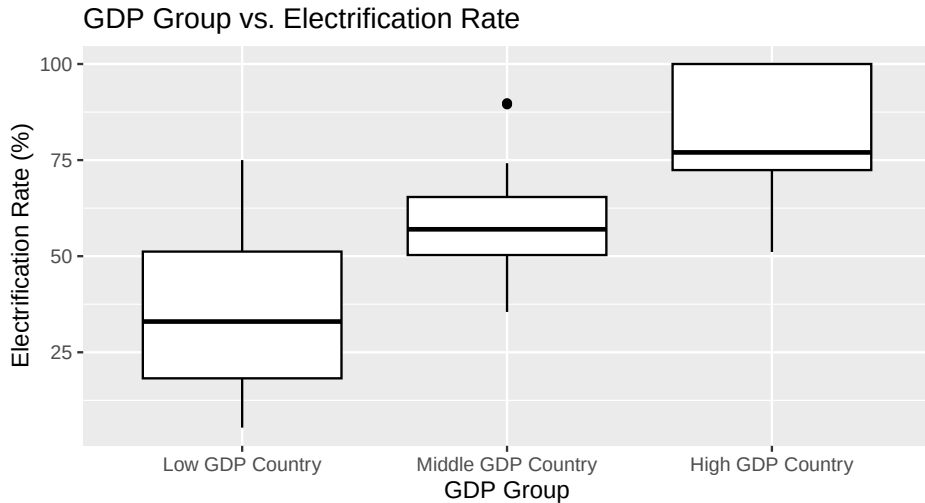
New variables/groups defined:

- **GDP Group** : countries were grouped into **Low, Middle, and High GDP** categories, with about one-third of countries in each group.

Approach:

- ① Clean data - remove implausible GDP and electrification values: Countries with GDP over \$50,000 and electrification rates over 100% are physically impossible.
- ② Compute mean and median electrification rate by gdp group
- ③ Visualize with a boxplot comparing electrification rates across GDP groups
- ④ Summarize in table showing electrification rate by GDP groups

Q1: Visualization



Source: CATF dataset.

Table 1: Electrification Rate by GDP per Capita Group

gdp_group	Count	Mean GDP	Median GDP	Mean	Median
		per Capita	per Capita	Electrification Rate (%)	Electrification Rate (%)
Low GDP Country	18	753.0	712.0	35.0	33
Middle GDP Country	17	1662.0	1695.0	60.0	57
High GDP Country	17	5794.3	4719.2	81.9	77

Q1: Insights & Limitations

- **Insights :** Economic development (GDP per capita) is **positively associated** with electrification rate across African countries: countries in the high-GDP group have much higher median electrification rates than those in the middle- and low-GDP groups. For CATF, this suggests that **economic development** is an important marker of electricity access.
- **Limitations:** GDP alone **cannot explain all** variation. Countries with the same GDP per capita, the electrification rate is not exactly the same. Future: add governance and infrastructure indicators.

Question #2 : (Eqraa) - Does the urban-rural electricity access gap differ across income groups?

Is the gap between urban and rural electricity access larger in poorer countries, and how does this equity divide shift as countries develop economically?

Relevance:

The urban-rural gap reveals whether access is equitably distributed within countries. Identifying which income groups have the largest gaps helps CATF prioritize off-grid and mini-grid interventions where rural populations are most underserved.

Q2: Variables & Approach

Variables: Urban Access Electricity Rate, Rural Access Electricity Rate, Income Group

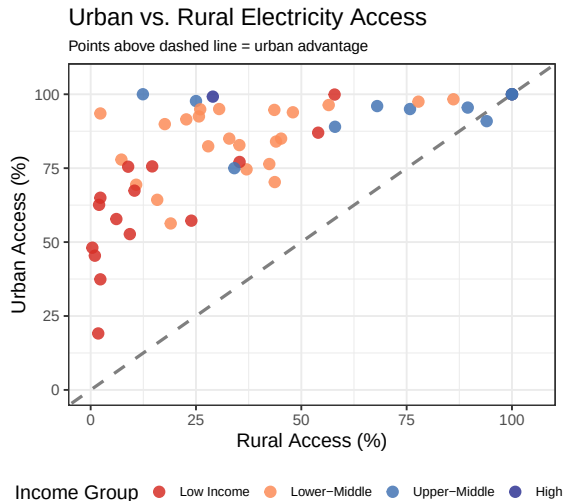
New variables:

- **Urban Rural Gap:** urban minus rural access rate (percentage points)
- **Income Group:** Low (less than \$1,000) / Lower-Middle (\$1,000 - \$4,000) / Upper-Middle (\$4,000 - \$12,500) / High (greater than \$12,500)

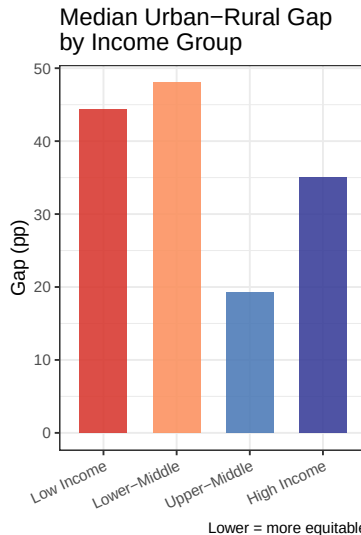
Approach:

- ➊ Removed countries with missing income group values or access rates below 0% or above 100%, as these are physically impossible
- ➋ Compute median urban-rural gap by income group
- ➌ Visualize with scatter plot and bar chart
- ➍ Summarize access rates and gap by income group in table

Q2: Visualization



Source: CATF dataset.



Q2: Summary Table

Table 2: Urban vs. Rural Access by Income Group

Income Group	n	Median Urban (%)	Median Rural (%)	Median Gap (pp)	Max Gap (pp)
Low Income	15	62.6	8.9	44.4	66.6
Lower-Middle	24	87.4	34.1	48.0	91.2
Upper-Middle	11	96.0	75.8	19.2	87.6
High Income	2	99.6	64.5	35.1	70.2

Q2: Insights & Limitations

- **Insights:** Every country has an urban advantage. Low-income median gap: 44pp vs. High-income: ~0pp. Gap narrows as income rises. For CATF, this suggests that rural access inequality is more severe in poorer settings, so the urban–rural gap may be a useful scorecard indicator for identifying underserved populations.
- **Limitations:** Urban/rural definitions vary across countries. The high-income group has a very small sample size, so comparisons for that group are less reliable. Future: standardize definitions and add reliability indicators.

Question #3: (Shutong) - Poverty Rate and Electricity Reliability

Does a higher poverty rate predict worse electricity reliability (SAIDI)?

Relevance:

Reliability is distinct from access – a country can have high electrification yet deliver unreliable power. Linking poverty to SAIDI helps CATF understand whether the poorest countries face a double burden of low access AND poor reliability.

Q3: Variables & Approach

Variables used:

- **Poverty Rate:** proportion below poverty line
- **SAIDI:** outage hours per customer per year

New variable:

- **Poverty Group:** countries were grouped into **Low, Middle, and High Poverty** categories, with about one-third of countries in each group.

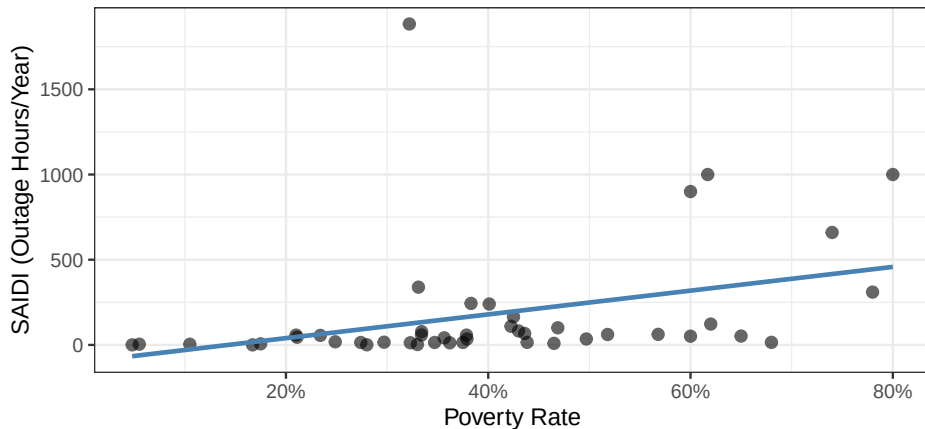
Approach:

- ➊ Removed poverty rates above 100%, as these are physically impossible
- ➋ Removed SAIDI values above 2,000 hours per year, as these are likely data errors
- ➌ Compute median **SAIDI** by **Poverty Group**
- ➍ Visualize with scatter plot and linear trend
- ➎ Summarize median and max **SAIDI** by poverty group in table

Q3: Visualization

Poverty Rate vs. Electricity Reliability (SAIDI)

Each point represents one African country



Source: CATF dataset. Lower SAIDI = more reliable.

Q3: Summary Table, Insights & Limitations

Table 3: Electricity Reliability (SAIDI) by Poverty Group

Poverty Group	n	Median Poverty Rate	Median SAIDI (hrs)	Max SAIDI (hrs)
Low Poverty	15	23.4%	12.8	1883.4
Middle Poverty	15	37.8%	67.2	339.0
High Poverty	15	60.0%	62.1	1000.0

- **Insights:** Higher poverty rates tend to be associated with worse electricity reliability, as countries with more poverty generally have higher SAIDI values. For CATF, this suggests that poverty may help identify where weak reliability is concentrated, making reliability an important scorecard indicator alongside access.
- **Limitations:** Only 10 -15 countries pass SAIDI quality filters. Also, SAIDI measures outage duration only and does not capture all dimensions of reliability. Future: expand collection; add SAIFI metrics.

Question #4: (Yiren) - Urbanization and Electricity Consumption

How does urbanization rate relate to electricity consumption per capita?

Relevance:

Consumption reflects how intensively electricity is used – a key indicator of economic activity and energy demand. Understanding how urbanization drives consumption helps CATF anticipate which countries face rapidly growing demand.

Q4: Variables & Approach

Variables used:

- **Urbanization Rate:** proportion of population in urban areas (0 -1 scale)
- **Electric Power Consumption:** kWh consumed per capita

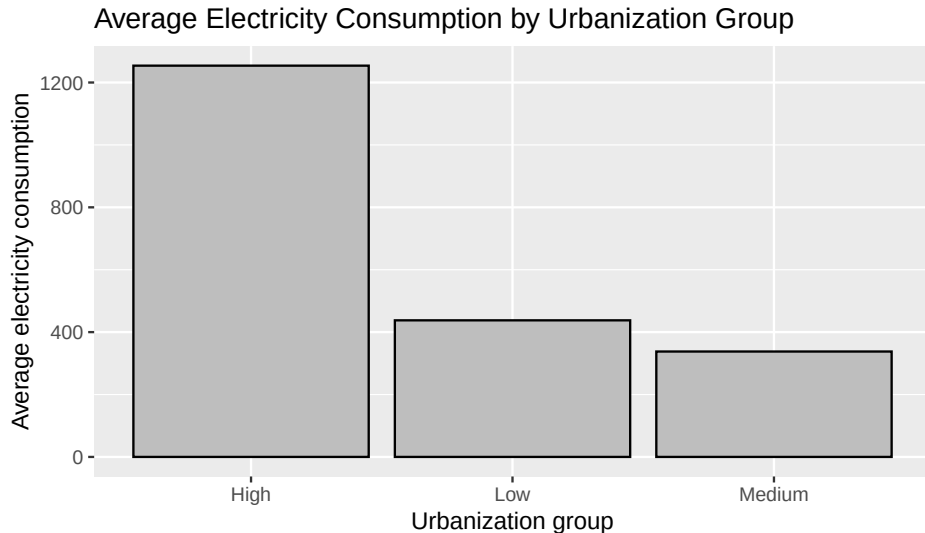
New variable:

- **Urban Group:** Low (less than 40%) / Medium (40-60%) / High (more than 60%)
urbanization rate

Approach:

- ① Filter missing urbanization and consumption values
- ② Convert **Urbanization Rate** to percentage for interpretability
- ③ Visualize with scatter plot and linear trend
- ④ Summarize mean and median consumption by **Urban Group** in table

Q4: Visualization



Q4: Summary Table, Insights & Limitations

Table 4: Electricity Consumption by Urbanization Group

Urbanization Group	n	Mean Consumption (kWh/capita)	Median Consumption (kWh/capita)
Low (<40%)	19	437.8	131.2
Medium (40-60%)	19	337.8	131.0
High (>60%)	14	1254.1	891.5

- **Insights :** Electricity consumption per capita is **much higher** in **highly urbanized countries**, while low- and medium-urbanization groups show similar consumption levels. For CATF, this suggests that urbanization may be most informative for identifying countries where electricity demand becomes substantially higher, which is relevant for planning and prioritizing sector indicators.

Q4: Summary Table, Insights & Limitations (Continued)

- **Limitations:** Urbanization alone does not fully explain differences in electricity consumption across countries, since demand may also depend on income, economic structure, and access. In addition, per capita consumption may miss informal or off-grid use. Future: add per-sector consumption data.

Overall Conclusion

Key Findings:

- Electrification rises sharply with GDP: 33% (Low) vs. 85% (High gdp group) - Q1
- Urban-rural gap exceeds 44pp in low-income countries – equity is not equal to access - Q2
- Higher poverty predicts worse SAIDI – showing reliability must be tracked separately from access - Q3
- More urbanized countries consume significantly more electricity per capita - Q4

Connection to Theme:

Economic development shapes not just whether people have electricity, but how equitably it is distributed, how reliably it is delivered, and how intensively it is consumed. The CATF Scorecard must capture access, equity, reliability, and consumption as distinct dimensions. This four dimensions should each be represented as distinct indicators in the CATF Scorecard.